Image quality

Code path:

```
~/dofbot_ws/src/dofbot_opencv/scripts/1.OpenCV_basic/03_quality_pic.ipynb
```

1. Compression method.

cv2.imwrite('yahboomTest.jpg', img, [cv2.IMWRITE_JPEG_QUALITY, 50])

cv2.CV_IMWRITE_JPEG_QUALITY: Set the image quality of the image format to .jpeg or .jpg, the value is 0---100 (the larger the value, the higher the quality), the default is 95

cv2.CV_IMWRITE_WEBP_QUALITY: Set the image quality of the image format to .webp format, the value is 0--100

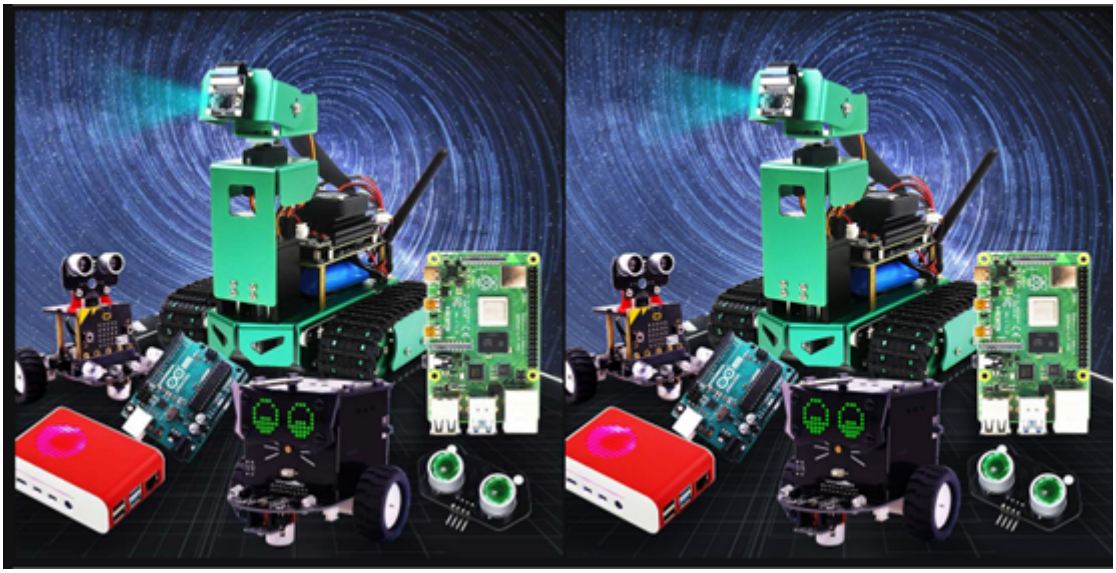
cv2.CV_IMWRITE_PNG_COMPRESSION: Set the compression ratio of the .png format, the value is 0--9 (the larger the value, the greater the compression ratio), the default is 3

The main code is as follows:

```
import cv2
img = cv2.imread('yahboom.jpg',1)
cv2.imwrite('yahboomTest.jpg',  img, [cv2.IMWRITE_JPEG_QUALITY, 50])
#1M 100k 10k 0-100 lossy compression
```

```
# 1 lossless 2 transparency attribute
import cv2
img = cv2.imread('yahboom.jpg',1)
cv2.imwrite('yahboomTest.png', img,   [cv2.IMWRITE_PNG_COMPRESSION,0])
# jpg 0 high compression ratio 0-100 png 0 low compression ratio 0-9
```

```
#bgr8 to jpeg format
import enum
import cv2
def bgr8_to_jpeg(value, quality=75):
      return bytes(cv2.imencode('.jpg', value)[1])
```

```
import ipywidgets.widgets as widgets
image_widget1 =   widgets.Image(format='jpg', )
image_widget2 = widgets.Image(format='jpg',   )
# create a horizontal box container   to place the image widget next to
eachother
image_container =   widgets.HBox([image_widget1, image_widget2])
# display the container in this   cell's output
display(image_container)
img1 = cv2.imread('yahboomTest.jpg',1)
img2 =   cv2.imread('yahboomTest.png',1)
image_widget1.value =   bgr8_to_jpeg(img1)
image_widget2.value =   bgr8_to_jpeg(img2)
```

When the code block runs to the end, you can see the comparison of the two photos.